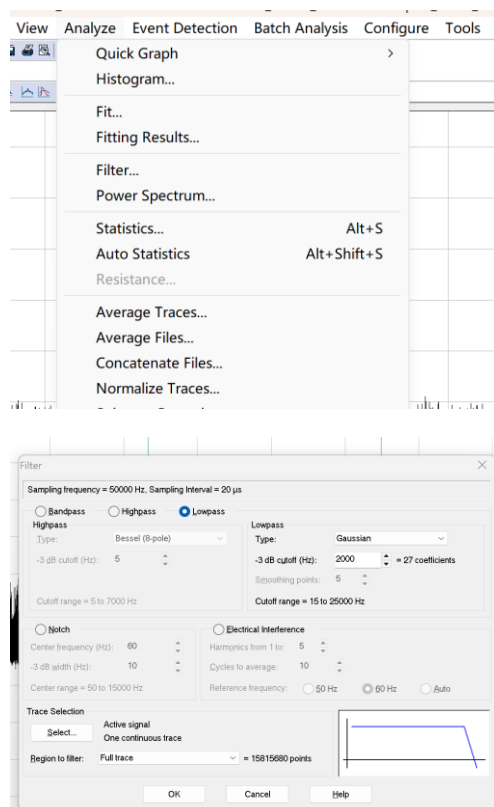


1. Filter:

Use Clampfit software to open “ABF” file, 'Click “Analyze”, choose “Filter”, then choose "Lowpass",
Type-“Gaussian”, 3dB cutoff (Hz) set as 2000 or 1000,
Region to filter: “Full trace” or cursors 1,2, depending on your data, then click “OK” to start filtering.



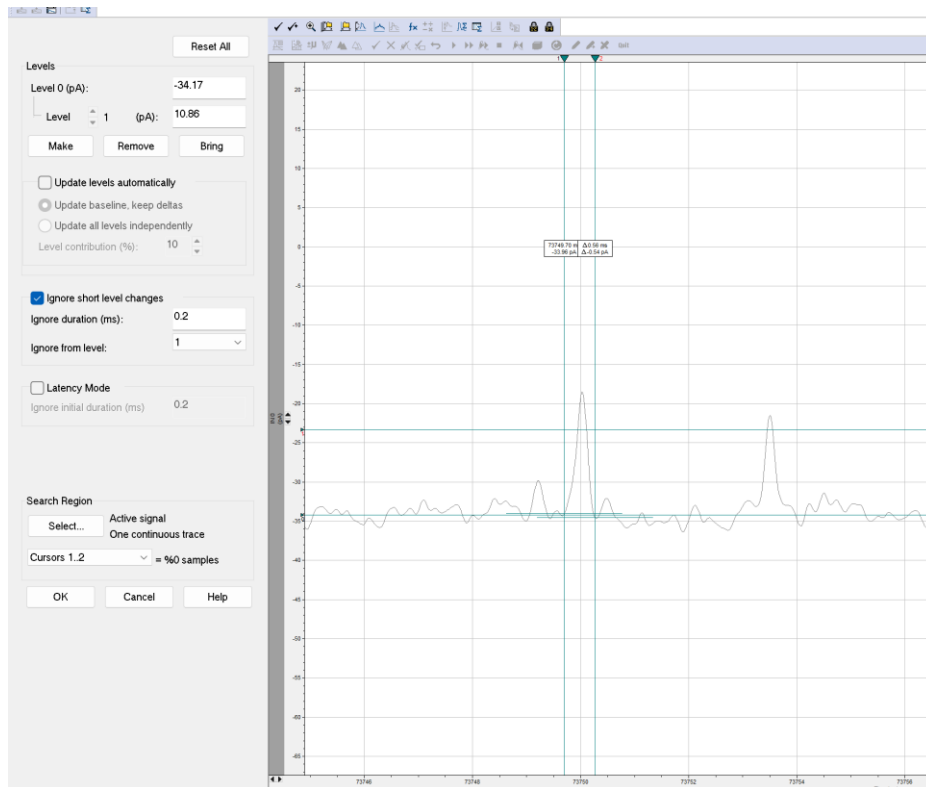
2. Set the Level 0 value for the open current and input the value into the origin (H1) in the Origin template file.

3. Single channel search

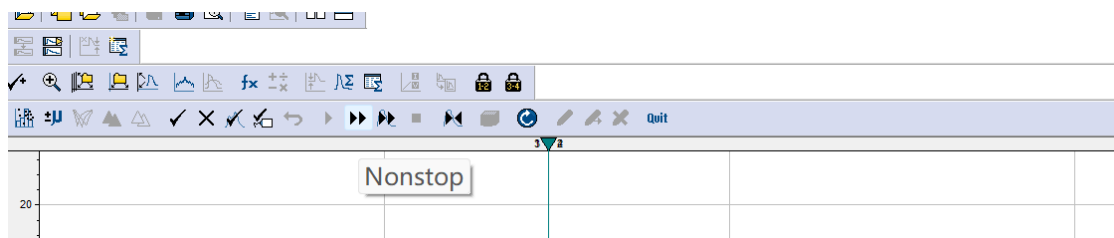
After getting the value of Level 0, set Level 1 depending on the current trace, but also need to consider the value of “sigma”. After setting level 0, untick “Update levels automatically”, if clicked, it may clash when searching the blockade events.

And for the ignore short level changes setting, the value of “Ignore duration (ms)” depends on the dwelling time of single blockade event, trying to find several represented blockade events showing a single molecule passing through and calculating the dwelling time in the current traces.

The time of “Ignore duration (ms)” should be lower than the calculated dwelling time, the example is shown in the figure below.



After setting all the parameters, search region can choose full trace or cursor 1, 2, which depends on the current traces, then click “OK” and “Nonstop” button to start “single channel search”.



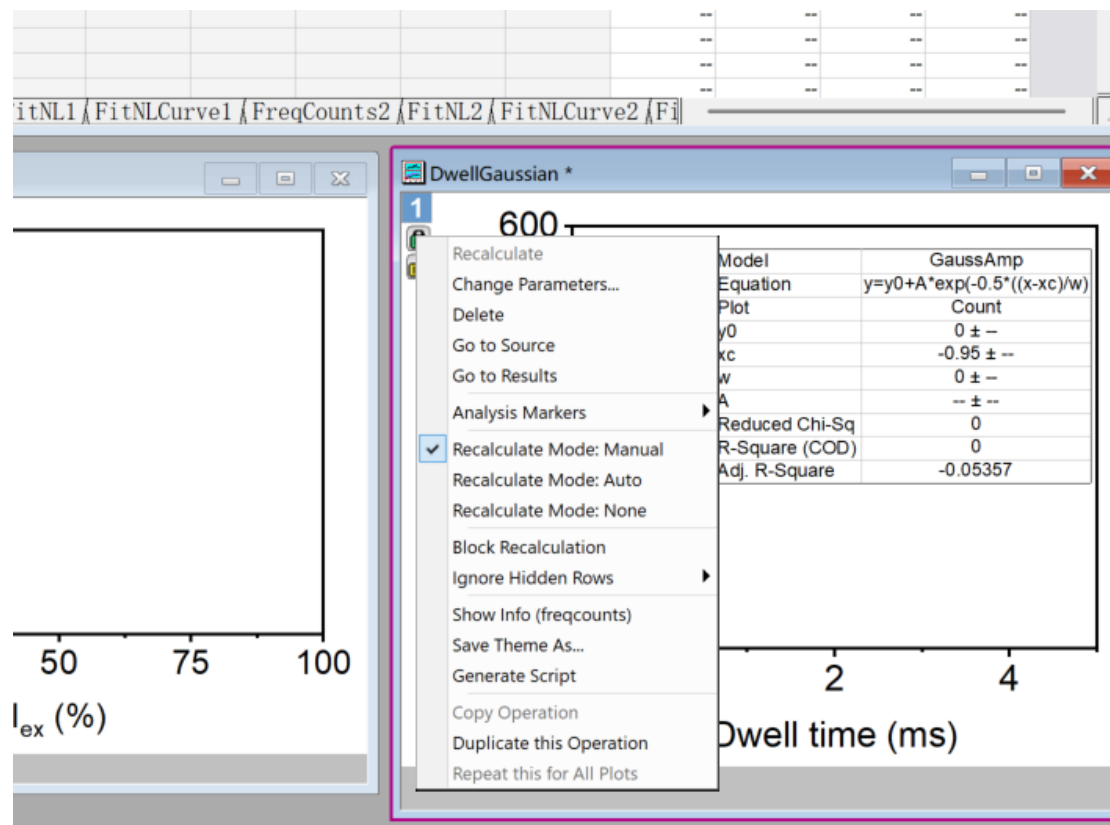
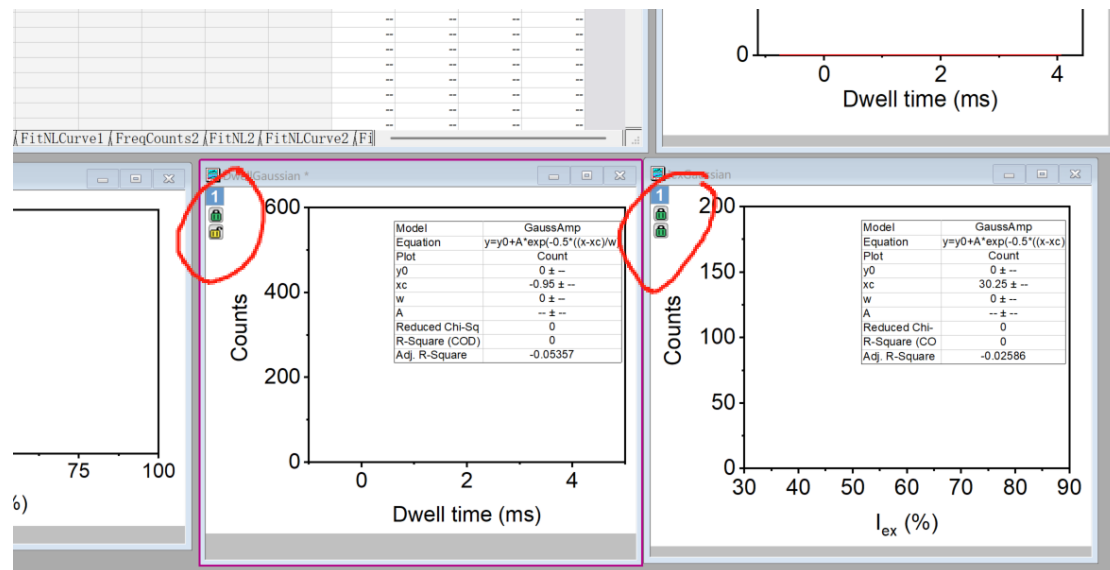
4. Copy data to the template origin file

After single channel search, click “window” and choose “Result 1:1” to get the data in “Events” Table. Copy the data from the “Level” to “Dwell time” to the Template “Peptide analyzer”, this template will automatically draw the “scatter”.

5. Calculate the dwell time and Iex% by using the Gaussian fitting

Click the first lock , click “change parameter” change fitting range, then click the second lock to recalculate.

Note: once the lock becomes yellow, it means the results need to recalculate.



6. In Origin, Frequency Count (frequency statistics) and Gaussian (Amp) fitting for a column of data can be accomplished through the following steps:

5.1. Perform "Frequency Count" on a column of data

Method One: Use menu operations

1. Select the data column (usually column Y)

2. Click **Statistics** → **Descriptive Statistics** → **Frequency Count...**
3. In the pop-up dialog box:
 - **Data Range:** make sure the data column is correct
 - **Bin Size:** The size of the interval can be customized (by default, it is calculated automatically)
 - **From Minimum / To Maximum:** Adjustable statistical range
4. Click **OK**, Origin will generate a worksheet(e.g. FreqCount1) , including:
 - **Bin Center**
 - **Counts**
 - **Sum**

5. 2. Perform Gaussian (Amp) fitting on the frequency statistics results

Step 1: Draw a histogram

1. In the newly generated frequency statistics worksheet (such as FreqCount1):
 - Choose **Bin Center** (X) and **Counts** (Y)
2. Click **Plot** → **Basic 2D** → **Column/Bar** → **Histogram** (or directly use **Ctrl + H**) .
3. Origin will generate a Histogram

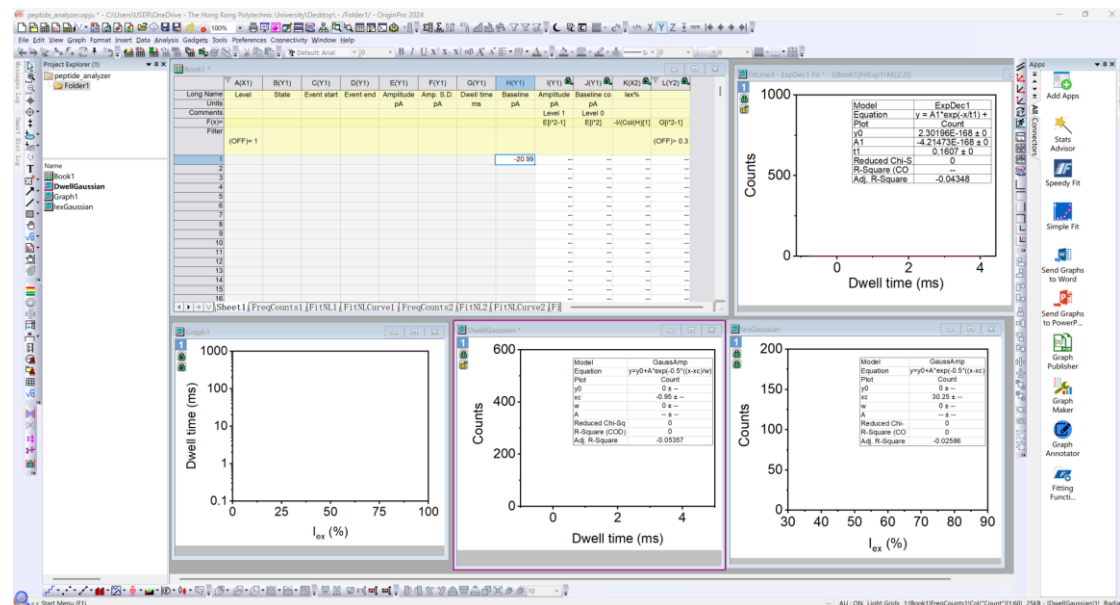
Step 2: Perform Gaussian (Amp) fitting

1. In the Histogram, right click the graph, → **Fit Peak** → **Gaussian (Amp)**
 - Or:
 - Click **Analysis** → **Fitting** → **Nonlinear Curve Fit** → **Open Dialog...**
 - In the pop-up dialog box:
 - **Function** choose **Gaussian (Amp)** (in the **Peak Functions** type) .
 - **Input** choose **Counts** as Y data, **Bin Center** set as X data.
2. Adjust the fitting parameters (such as the initial guessed values), and click **Fit**
3. Origin will generate the fitting results, including:
 - Fitting curve (red lines superimposed on the histogram)

- **Parameter table** (e.g. **Amplitude**、**Center**、**Width**. etc)
- **Goodness of fit** (R^2 、 χ^2 , etc)

Optional: Adjust the fitting range

- In the Nonlinear Curve Fit dialog box, you can manually adjust From and To to limit the fitting range.



Multiple peak fit:

First, perform a Frequency Count on the column of data that needs to be fitted.

Then select the statistical graph,

click Analysis-Peaks and Baseline-Multiple Peak Fit-Open Dialog, confirm that the Input value is the data value that needs fitting, and select Peak Function-Gauss Amp-OK;

Use the cross cursor, double-click to select the approximate position of the peak top of the first peak, then double-click to select the peak top position of the second peak.

After that, click Open NLFit to confirm the position of the selected peak, that is, the value of X_c , and then click Fit.

